

(a) Find, using calculus, the x coordinates of the stationary points of C .

(4)

(b) Hence find the values of x for which y is increasing.

(2)

- (b) Find, according to these models, the total amount of money that would be given to local charity by the adult population of the town from 2019 to 2032 inclusive. Give your answer to the nearest £1 000
- (3)**

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(c) Using algebra and showing each step of your working, fully factorise $f(x)$. (5)

[illegible]

5. (a) Given $0 < a < 1$, sketch the curve with equation

$$y = a^x$$

showing the coordinates of the point at which the curve crosses the y -axis.

(2)

x	2	2.5	3	3.5	4
y	4.25	6.427	9.125	12.34	16.06

The table above shows corresponding values of x and y for $y = x^2 + \left(\frac{1}{2}\right)^x$

The values of y are given to 4 significant figures as appropriate.

Using the trapezium rule with all the values of y in the given table,

- (b) obtain an estimate for $\int_2^4 \left(x^2 + \left(\frac{1}{2}\right)^x \right) dx$

(3)

Using your answer to part (b) and making your method clear, estimate

- (c) $\int_2^4 \left(x(x-3) + \left(\frac{1}{2}\right)^x \right) dx$

(2)

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7. Given $\log_a b = k$, find, in simplest form in terms of k ,

$$(i) \log_a \left(\frac{\sqrt{a}}{b} \right) \quad (2)$$

$$(ii) \frac{\log_a a^2 b}{\log_a b^3} \quad (2)$$

$$(iii) \sum_{n=1}^{50} (k + \log_a b^n) \quad (3)$$

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